

Developing a finite WRRRA toy model to explore quantum gravity vacuum energy and dark sector connections

Wonsik Choi · Jeongin Choi

Independent Research, Seoul, Republic of Korea

Correspondence: janefather@gmail.com · ORCID 0009-0001-4263-9772

Paper v1.8 · 8 October 2026

Abstract

We develop a finite Worldline–Residue–Resource–Action (WRRRA) toy model to explore possible explanatory connections among quantum states and gravity, vacuum energy and expansion, and dark-matter and dark-energy effects. The purpose is model development and assessment of specified computational connections, rather than a claim to have solved these three problems or established WRRRA as the physical theory of the universe. The guiding hypothesis is that one nonprivileged universe can be described through minimal computation, a common carrier and phenotype. The implemented model maps a finite SOURCE and conditional carrier states through sector admission to information loads, calibrated energy, pressure and downstream gravity and expansion responses. Verified constants and observed sector shares are disclosed construction inputs. We identify what follows from those inputs and the WRRRA transformations, what is reproduced, and where the proposed connections remain incomplete. In a model with 1,014,999 active addresses and a 128-dimensional carrier, exact correlation-preserving reduction reproduces loads to 8.9×10^{-16} . Identical separate marginals need not preserve joint loads or calculated deceleration. A class-resolved lift repairs a concrete controlled-update failure. Applying the frozen WRRRA acceleration scale with the constitutive coefficient $\kappa=1/8$ through the credited empirical RAR kernel to 2,693 published SPARC points gives 0.13288 dex RMS without a new fit. These results substantiate the toy model’s interface, identify its dynamical and observational limits, and document descriptive consistency. They do not derive microscopic quantum gravity or the small cosmological constant. The contribution is an executable, auditable model of a shared explanatory chain, including a precise account of which microscopic distinctions its present macroscopic outputs cannot resolve.

Keywords: WRRRA; finite toy model; model development; common carrier; quantum gravity; vacuum energy; dark sectors

1. Model development and the three explanatory questions

This study develops a finite toy model for investigating possible connections among three unresolved questions: quantum states and gravity, vacuum energy and expansion, and dark-sector effects. A toy model is a deliberately restricted construction in which assumptions, transformations and outputs can be inspected and tested. The research task is to implement and assess a proposed explanatory mechanism within that construction. It does not require that the complete physical questions have already been resolved.

The WRRRA hypothesis organizes one nonprivileged universe through minimal computation, a common carrier and phenotype. The model implements SOURCE → conditional carrier states → sector admission and phenotype → information loads → calibrated energy and pressure → gravity and expansion responses. The shared organization is the proposed explanatory structure. The calculations below determine its consistency, dependencies and present explanatory reach; they do not establish its unique physical realization.

The specific contribution is a worked WRRRA interface result: replacing the correlated address–carrier state by its separate marginals can change the modeled pressure and deceleration, whereas retaining three weighted operators preserves every stipulated sector carrier readout. That snapshot representation nevertheless fails for the actual mod-3 controlled update; an explicit rank-six class lift restores closure. The model therefore identifies which information may be discarded, which correlations affect its gravity readouts, and which additional information its specified updates require. Calibration and inherited response laws are recorded separately from these consequences. The three physical questions motivate the construction; the results below assess its bounded explanatory mechanism.

Here “SOURCE” means the finite address measure and admission rule in Section 2; “phenotype” means the P-sector admission, not a derived catalogue of particle species. “Minimal computation” means preserving the declared readout/update contract while comparing information and implementation costs; it does not assert a global physical minimum. “Nonprivileged” is the motivating universality hypothesis, not an empirically established property of this particular arithmetic construction.

Observable-algebra reduction [2] organizes a state description around a restricted observable set. Exact dynamical reduction [3] additionally addresses preservation under a specified evolution. This work applies those distinctions to the disclosed WRRR coefficients: the concrete contribution is the actual-interface counterexample, its smallest linear class lift for the complete class-resolved arbitrary-carrier query family, and the resulting admissible macro domain. The general reduction identities are established mathematics. Their WRRR application makes a model-development decision testable: a snapshot cache cannot be used as the full state of this controlled evolution.

Evaluation of the three proposed connections

Quantum states and gravity. Verified inputs: normalized carrier states, disclosed operators and calibrated physical coefficients. WRRR transformation: the joint address–carrier state is mapped to sector-weighted carrier operators, loads, energy and the downstream gravity response. Outputs: exact preservation of specified correlated readouts and calculated expansion, rotation and conditional lensing; identical marginals can give different loads and deceleration. Falsification conditions: loss of state positivity or normalization, disagreement between direct and reduced readouts, failure of shared-energy bookkeeping, or disagreement with specified physical observations. This is an implemented connection between quantum state information and gravity readouts. It is not yet an internally calculated microscopic quantum-gravity theory; heterotic-string support is not counted as internal integration without the corresponding WRRR calculation.

Vacuum energy and cosmic expansion. Verified inputs: reference energy density, sector shares and disclosed dimensional calibrations. WRRR transformation: finite SOURCE admission and resident/return loads feed a finite energy function whose volume dependence determines pressure and expansion. Outputs: finite sector energy and pressure and a computed deceleration. Falsification conditions: inconsistent units, nonconserved exchange, disagreement between pressure and the stated energy derivative, or failure of the fixed model to reproduce the specified expansion observations. The finite construction avoids an unbounded ledger, but the observed small energy scale currently enters through calibration; finiteness by itself is not a derivation of that scale or a completed solution of the cosmological-constant problem.

Dark matter and dark energy. Verified inputs: reference shares 5%, 26.8% and 68.2%, source parameters and published acceleration measurements. WRRR transformation: arithmetic admission separates phenotype, resident nonphenotype and return sectors; their loads enter the shared energy and twist/local-gravity response. Outputs: reproduction of the calibrated shares and calculated expansion, rotation and conditional lensing, with a descriptive published-data acceleration comparison. Falsification conditions: violation of the common ledger, separately refitted outputs that no longer follow the same fixed state, or failure against specified rotation, lensing or expansion data. The RAR function is inherited and credited; the comparison is not a complete lensing or cosmological-data validation.

Taken together, these are a toy-model proposal whose specified connections are assessed by the computations below. They are not presented as three completed resolutions. Each output is assessed through its disclosed transformation and failure conditions.

2. Verified inputs and the finite construction

Let the active address set be $n=2, \dots, N$, with $N=1,015,000$. Address 1 is inactive. Each address carries a normalized d -dimensional density operator ρ_n . Its classical weight is $w_n > 0$ with sum one. The complete classical–quantum state is

$$\Omega = \sum_{n=2}^N w_n |n\rangle \langle n| \otimes \rho_n, \rho_n \geq 0, \text{Tr } \rho_n = 1.$$

This is a classical address register coupled to quantum carrier blocks; coherence between addresses is not part of this contract. Carrier coherence is allowed. In the numerical construction, $d=128$. Sector labels P,D,R denote phenotype, resident nonphenotype and return. Nonnegative admission effects $e_s(n)$ sum to one at every address. Define

$f_s(n)=e_s(n)g_s(n)$, where g_s is a disclosed positive response. The general results below need no particular arithmetic classifier. The numerical instance uses the following rules.

$$w_n = \frac{n^{-\alpha}}{\sum_{m=2}^N m^{-\alpha}}, g_s(n) = 1 + \lambda_s \frac{\log n}{\log N}, (\lambda_P, \lambda_D, \lambda_R) = (0, 0.25, 0.1).$$

Primes have $(e_P, e_D, e_R) = (0, 0, 1)$; even composites have $(0, 1, 0)$. Odd composites use a calibrated four-mode, eight-frame admission:

$$d_k(n) = \sum_{j=1}^4 \frac{1}{2} \cos[y_j(\log n + \xi k)], t_k(n) = \frac{1}{1 + e^{-(d_k(n) - h)}},$$

$$e_P(n) = 1 - \prod_{k=0}^7 (1 - t_k(n)), e_D(n) = 0, e_R(n) = 1 - e_P(n).$$

The gamma values are 14.134725141734695, 21.022039638771556, 25.010857580145690 and 30.424876125859512. These are adopted spectral inputs; four-mode truncation is specified rather than physically derived. The common carrier is a periodic 128-site lattice. With S the cyclic shift and $D_0 = |0\rangle\langle 0|$, define

$$K_c = 2I - S - S^\dagger, K_b = \frac{K_c + \epsilon D_0}{1 + \epsilon / (2d)}, (A_P, A_D, A_R) = (I, K_c/2, K_b/2).$$

The denominator normalizes the trace of K_b to $2d$. These operators are nonnegative; they need not be bounded by I . The diagnostic $\epsilon=0$ and 8 changes the supplied carrier operator, without refitting dimensional coefficients.

Input	Value or rule	Status and role
Sector targets	0.05, 0.268, 0.682	Reference inputs; only two independent after normalization
N	$\text{lcm}(20, 250, 500) \times 29 \times 70$	Finite candidate based on rounded targets; no physical quantization law
α	1.8996877935161325	Joint calibration to address shares
h	1.4476744689338703	Joint calibration to address shares
ξ ; frames	0.1; 8	Specified admission rule
d ; ϵ	128; 0 or 8	Carrier size and diagnostic perturbation
η_P	$7.5615824991 \times 10^{-10} \text{ J/m}^3$	Frozen reference-state SI calibration
η_D	$7.2857613288 \times 10^{-10} \text{ J/m}^3$	Frozen reference-state SI calibration
η_R	$7.6461028188 \times 10^{-10} \text{ J/m}^3$	Frozen reference-state SI calibration
V_0 ; n_P, n_D, n_R	1 m^3 ; 0, 0, 3	Reference volume and constitutive volume exponents
H_0 ; u_{crit}	67.4 km/s/Mpc; $7.6689477678 \times 10^{-10} \text{ J/m}^3$	Inherited reference constants
Local diagnostic	Plummer $M=6 \times 10^{10} M_{\text{sun}}$, $b_0=3 \text{ kpc}$	Specified source; not inferred galaxy profile
RAR comparison	Official RAR.mrt, 2,693 rows	Published observations; no fit in this revision

The rounded-target construction is not robust evidence for a unique capacity. For example, transferring 0.0001 from D to R changes the rational denominator and the candidate N to 20,300,000. Earlier frozen-response probes of large cutoffs are included in the reproduction package [7]. The model therefore retains a finite candidate and sensitivity analysis, not an observed measurement of the universe's storage size. Pure address and SI-energy fractions also remain distinct ledgers: equal address weights do not imply equal physical energies.

Reference state and coefficient provenance

The address-share targets (0.05, 0.268, 0.682) and the inherited SI-energy targets (0.0493, 0.265, 0.6857) are distinct inputs. In `snapshots/source_step2/dependencies/original/m10_results.json`, the frozen 0.9 address measure with uniform carrier I/d has reference moments (0.05000000000000009, 0.2789373775449565, 0.687748727555431). Since $\text{Tr}[(I/d)A_s]=1$, the original calibration is $\eta_s = u_{\text{crit}} F_s / \mu_s(\text{ref})$, at $a=1$ and $V_0=1 \text{ m}^3$. This reproduces the η values above. Subsequent arithmetic and correlation probes retain these coefficients; their changed moments need not reproduce the original energy fractions exactly. Thus 5/26.8/68.2 is an address-admission calibration, not a second assertion about the same energy ledger.

$H_0=67.4 \text{ km/s/Mpc}$ is the adopted Planck 2018 base-Lambda-CDM reference [8]; $u_{\text{crit}}=3 H_0^2 c^2/(8 \pi G)$ is calculated from that anchor. The historical energy fractions are the project's frozen rounded reference convention, not claimed here as the exact entries of a newly fitted Planck likelihood. The address targets are the project's separate rounded construction targets. The frame offset is ξ_k with $\xi_i=0.1$ and $k=0, \dots, 7$; no independent ξ_k parameter is introduced.

3. Exact correlation-preserving interface

Define the three unnormalized sector-weighted carrier operators and scalar loads:

$$\Sigma_s = \sum_n w_n f_s(n) \rho_n, L_s = \text{Tr}(\Sigma_s A_s).$$

Proposition 1 (readout sufficiency and positive realization). For fixed f_s , each Σ_s is nonnegative and exactly preserves every sector carrier query $\text{Tr}(\Sigma_s B)$, for arbitrary Hermitian B . The collection admits a completely positive trace-preserving realization when a failure branch is retained.

Proof. Positivity follows from nonnegative coefficients. Linearity of trace gives the query identity. Choose $M \geq \max_n \sum_s f_s(n)$. On the address-classical input, Kraus operators $J_{\{s,n\}} = \sqrt{f_s(n)/M} |s\rangle\langle n| \otimes I$ produce the flagged blocks Σ_s/M . Add $J_{\{F,n\}} = \sqrt{(1 - \sum_s f_s(n)/M)} |F\rangle\langle n| \otimes I$. Their adjoint products sum to the identity. The resulting state has unit trace and is a valid quantum channel output. Multiplying the unnormalized flagged sector block by M recovers its weighted operator. If the carrier state is normalized after postselection, its sector probability must also be retained. Σ_s itself is an unnormalized bookkeeping operator, not a density matrix or an extra energy sector. ■

For G diagnostic classes with common conditional states τ_k and address class $k(n)$, one can stream

$$B_{sk} = \sum_{n: k(n)=k} w_n f_s(n), \Sigma_s = \sum_{k=1}^G B_{sk} \tau_k.$$

This keeps correlations between sector response and carrier state. It does not replace the joint state by separate marginals. The grouping is a test input, not a derivation that all physical addresses fall into these classes.

Proposition 2 (restricted linear dimension). Let $F_{\{sn\}} = f_s(n)$ have rank r and fix strictly positive w_n . On the interior of the space of trace-one conditional carrier blocks, all-sector arbitrary-carrier readout variations have real dimension $r(d^2-1)$. An exact linear encoding for that entire query family requires at least this many variable real coordinates; r independent weighted operators attain the bound.

Proof. Expand each perturbation δp_n in a traceless Hermitian basis with d^2-1 elements. In each basis direction, $\delta \Sigma = F \text{diag}(w) \delta x$. Since $\text{diag}(w)$ is invertible, its image has dimension r . Directions are independent; sufficiently small perturbations around I/d preserve positivity. Thus the total image dimension is $r(d^2-1)$. A lower-dimensional linear encoding has a nonzero kernel that changes a target expectation; an appropriate Hermitian query separates that direction. Choosing a basis of the row space of F attains the dimension, with known fixed trace components. ■

This is a bound for a fixed query family and linear encodings. For three fixed A_s at one snapshot, just three scalar L_s suffice. Neither bound constrains nonlinear physical implementations or proves the physical necessity of the chosen arithmetic law. The dimension argument is standard linear algebra; its role is to define precisely what the WRR compression preserves.

4. Product-marginal error and dynamical closure

Let $\mu_s = \sum_n w_n f_s(n)$, the carrier marginal be $\rho_{\text{bar}} = \sum_n w_n \rho_n$, and $b_s(n) = \text{Tr}(\rho_n A_s)$. The product approximation is $L_s^{\text{prod}} = \mu_s \text{Tr}(\rho_{\text{bar}} A_s)$. Its error is exactly

$$L_s - L_s^{\text{prod}} = \text{Cov}_w(f_s, b_s), \left| L_s - L_s^{\text{prod}} \right| \leq \frac{a_{s,\text{max}} - a_{s,\text{min}}}{2} \sqrt{\text{Var}_w(f_s)}.$$

The identity follows by subtracting the product of the two means. Cauchy–Schwarz bounds the covariance, and a random variable in the spectral interval of A_s has variance at most one quarter of that interval’s squared width. This bound can be loose; the benchmark records both the observed error and bound. Zero spectral width makes the error zero, explaining preservation of the phenotype identity budget even when D or R loads change.

Snapshot sufficiency need not imply closed dynamics. A common carrier channel Φ independent of n obeys $\Sigma'_s = \Phi(\Sigma_s)$, so weighted operators remain sufficient. Address-controlled channels instead act before the weighted sum and generally require additional moments. In an equal-weight two-address qubit example, states $(|0\rangle\langle 0|, |1\rangle\langle 1|)$ and the swapped pair have the same average $I/2$ and the same initial Z expectation zero. Applying I at the first address and X at the second yields average $Z = +1$ and -1 , respectively. A marginal-only reduced state cannot replay this update.

Thus the correct contract is: fixed query operators at a snapshot, or a disclosed common carrier channel. An arbitrary new address-dependent update requires a fresh sufficiency or invariant-space check [3]. This prevents a compressed state from being described as a universally closed microscopic simulator. In the two-address example take all f_s constant: all three weighted operators are then equal between the two initial states. A stronger counterexample below uses the actual WRRRA coefficients.

Proposition 3 (class-controlled lift). Partition addresses into fixed disjoint classes C_k , keep w and f fixed, and let class k evolve by its disclosed common carrier channel Φ_k . Define

$$\Sigma_{sk} = \sum_{n \in C_k} w_n f_s(n) \rho_n, \Sigma'_s = \sum_k \Phi_k(\Sigma_{sk}).$$

These class-resolved operators replay such updates exactly, and each block itself evolves as $\Phi_k(\Sigma_{sk})$. For the complete class-resolved carrier-query family, Proposition 2 gives minimum variable linear dimension $r_{\text{lift}}(d^2-1)$, where r_{lift} is the rank of the rows $f_s(n) 1_{\{C_k\}}(n)$. This is sufficiency and a linear bound for that complete family; it is not a claim that every specified channel requires every class-resolved query. Proof: apply each linear channel inside its class sum and apply Proposition 2 to the lifted coefficient matrix. ■

Closure criterion and reusable audit. For arbitrary tuples of common CPTP maps on the specified classes, the original weighted interface is closed if the row space of F is invariant under multiplication by each class indicator. Equivalently, stacking the class-masked rows does not increase its rank. Then each Σ_{sk} is a linear combination of the original operators, so Proposition 3 supplies the update. Conversely, if a masked row leaves this row space, project it orthogonally onto the original null space. This yields a nonzero perturbation invisible to the original interface and visible to that class. Opposite small traceless carrier perturbations preserve positivity; an identity/bit-flip controlled update can distinguish the two states. Thus universal closure for this specified channel family cannot be asserted. This is an application of established invariant-space reduction [2,3], not a new general sufficiency theorem.

The audit has five reusable steps: (i) state the query family and form its address coefficient matrix F ; (ii) compute its rank and the retained operator family; (iii) mask its rows by the proposed control classes and compare the lifted rank; (iv) when rank grows, construct a null-space counterexample and retain class-resolved blocks; (v) choose a representation by measuring construction, storage and the required query workload. It applies to finite classical–quantum mixtures beyond arithmetic addresses. A snapshot cache is useful only when its contract matches the computation to be performed.

For the actual WRRRA f_s , use $C_1 = \{n: n \bmod 3 = 0\}$, C_2 its complement. The initial rank is three, while the lifted rank is six. For $d=128$ the complete class-resolved family has 98,298 variable coordinates, versus 49,149 for the original three-operator family. An exact rank witness uses addresses 3,6,9 in the divisible-by-three class and 2,4,25 in its complement. In each class the prime, even-composite and odd-composite columns give a triangular 3×3 minor with

strictly positive R, D and P pivots. The disjoint class supports combine these minors into rank six; no floating-point rank decision is needed. This answers a concrete implementation question: three scalar loads suffice for frozen cosmological outputs; three operators support changing snapshot carrier queries; class-resolved operators support the specified address-controlled updates. These are different information contracts, not interchangeable definitions of a universal minimum.

5. Shared energy, records and observable equivalence

The supplied energy and volume laws are

$$E_s(a) = \eta_s V_0 a^{n_s} L_s, V = V_0 a^3, \rho_s = E_s / V, P_s = -n_s \rho_s / 3.$$

They give pressureless P,D and constant-density R for frozen L. This matter-plus-constant-density behavior is a constitutive input, not a new derivation of accelerated expansion. Pressure is the derivative of the same E used for the density readout, at fixed L. Internal state evolution requires its own disclosed generator; this derivative is not the total derivative along evolving L.

Recording is checked in a boundary that includes its finite supplier. The same work cannot be counted both as new cosmic energy and as an already allocated phenotype resource:

$$E_{sys,before} + E_{supplier,before} = E_{sys,after} + E_{record,after} + E_{supplier,after}.$$

Here the record starts blank, its final energy is displayed separately, and supplier work equals the system-energy change plus record energy. The inherited effective-record, magnetic internal and joint internal models are alternatives, not simultaneously added particle species. The frozen supplier branches and their numerical budgets are documented in Appendix B. Let $\rho = \text{sum_sp_s}$ and $P = \text{sum_sP_s}$. The homogeneous and local diagnostic readouts use

$$H / H_0 = \sqrt{\rho / u_{crit}}, q = 1/2 (1 + 3 P / \rho), a_T = c H_0 \sqrt{\rho_D / (8 u_{crit})}.$$

The local rule is $g(r) = g_b(r) v(g_b/a_T)$, $v^2 = rg$, with $v(y) = 1/[1 - \exp(-\sqrt{y})]$. This is the existing empirical RAR function of McGaugh, Lelli and Schombert [4], evaluated here at the state-derived a_T . We do not claim the function itself is a WRA discovery. Conditional lensing integrates this same g under the stipulated weak-field condition $\Phi = \Psi$. Neither this local prescription nor the homogeneous relation supplies a complete four-dimensional gravitational theory.

Proposition 4 (three-load quotient and inverse within the renderer). With coefficients, volume, local source and response kernel fixed, any two microscopic states with equal L_P, L_D, L_R have identical H, q, a_T, v and conditional lensing. At positive density, the three densities are uniquely determined by H, q, a_T :

$$\rho = u_{crit} (H / H_0)^2, \rho_R = \frac{1 - 2q}{3} \rho, \rho_D = 8 u_{crit} \left(\frac{a_T}{c H_0} \right)^2, \rho_P = \rho - \rho_R - \rho_D.$$

Proof. The forward map factors through the three loads, establishing equivalence. Since $P = -\rho_R$, the q relation yields the second inverse; the a_T relation yields the third. Remaining density follows by subtraction. At fixed r and positive g_b , $g_b/[1 - \exp(-\sqrt{g_b/a_T})]$ increases strictly with a_T , so a rotation value in its allowed range also determines a_T . Nonnegative recovered densities define the unrestricted scalar renderer domain. They are necessary but not sufficient to belong to the image of the fixed microscopic construction. ■

For fixed w, f and scale factor, $A_P = I$ fixes $L_P = \mu_P = \sum w_{nf_P}(n) = 0.05$ independently of the conditional carrier states. Consequently the actual fixed-SOURCE macroscopic image has at most two variable loads. In this arithmetic instance, $f_D f_R = 0$ at every address: D and R occupy disjoint supports. With otherwise arbitrary normalized conditional states, their exact attainable domain is

$$L_P = \mu_P.$$

$$\mu_D \lambda_{min}(A_D) \leq L_D \leq \mu_D \lambda_{max}(A_D).$$

$$\mu_R \lambda_{min}(A_R) \leq L_R \leq \mu_R \lambda_{max}(A_R).$$

Necessity follows from the spectral expectation bounds. Sufficiency follows by assigning independent extremal-state convex mixtures on the disjoint D and R supports. This rectangle is for the full conditional-state family, not for every constrained 32-group diagnostic family. At fixed a, convert the recovered densities to loads using the disclosed η and volume law and test this rectangle, including the fixed P equality. With $\epsilon=0$, both D and R operators range from zero to two; $\epsilon=8$ changes the R interval. The archive records these bounds and checks the reported states against them.

The unrestricted three-density renderer has information dimension at most three; its fixed-SOURCE microscopic image has at most two variable coordinates. A simpler three-load model is an exact baseline for these outputs. SOURCE structure is resolved by conditional address/carrier queries, not by declaring these already-equivalent macro numbers to be independent microscopic evidence. This equivalence is an explanatory constraint on what the present model can claim.

What the coupling explains within this model

Verified inputs are the same address marginal, the same carrier marginal, positive conditional states, and frozen eta and volume exponents. The WRRRA transformation retains the covariance between sector response $f_s(n)$ and carrier expectation, passes its load changes into the same energy, and uses $P=-\rho_R$ in the common homogeneous response. The output is $q=-0.52856$ for the product replacement and $q=-0.54501$ with correlation retained (Section 6). This demonstrates a specific mechanism inside the toy model: an address-carrier correlation invisible to both separate marginals changes the return-sector density and hence the pressure-to-density ratio. It explains why marginal-only state descriptions are insufficient for the proposed gravity interface. It does not establish that the tested correlation is the measured cosmic state. The implementation fails if direct and retained-joint calculations disagree beyond tolerance or the covariance identity fails; observational adequacy is a separate empirical question.

A consequential constraint also follows from sharing the same ledger. For fixed SOURCE and scale factor, define $p=\rho_P/u_{crit}$, $h=H/H_0$ and $t=a_T/(c H_0)$. Then

$$(1+q)h^2 = \frac{3}{2}(p+8t^2).$$

It follows by eliminating ρ_R from q and ρ_D from a_T . The 12 archived shared states satisfy this identity to 4.45×10^{-16} . Once p, h and t are fixed, q cannot be fitted independently; at fixed h , increasing the clustering contribution forces q upward under this law. This is a conditional consistency relation of the stipulated shared-energy response, not a newly derived gravitational field equation. Failure against a jointly specified physical state would reject that fixed response combination; introducing unrelated output-specific coefficients would change the model being tested.

Alternative constructions with the same calibrated shares

We bring the existing controlled architecture comparison into the main statement. All alternatives retain alpha and the SI coefficients and legitimately calibrate their admission threshold (or constant beta) to the same address shares. Define $v_3(n)$ as the exponent of 3 in n and Q as the P-conditioned weight of addresses with odd $v_3(n)$. This is a conditional component-address query, not an observed particle abundance.

Admission	Q	q at $a=1$
One frame	0.4195836874	-0.5285597845
Eight frames	0.4020461347	-0.5285585894
Constant admission	0.3998832014	-0.5285589334

The same aggregate calibration does not make these conditional explanations identical. Eight frames and constant admission differ in Q by 0.0021629333, while their q values differ by only 3.44×10^{-7} and their current rotation outputs coincide. The exact eight-frame recurrence preserves Q and every address admission while reducing the direct trigonometric evaluations. This comparison supports retaining the phase architecture when the declared task includes that address/frame explanation; it does not empirically select eight frames. If only the macro outputs are requested, the load quotient remains the adequate smaller description.

[snapshots/source_step3/structure_comparison.py](#) provides the construction and results.

Acceleration coefficient and sensitivity

Write $a_T = c H_0 \sqrt{\kappa \rho_D / u_{\text{crit}}}$. The inherited $\kappa = 1/8$ is present in `snapshots/source_step2/connection_sources/carrier_compute.py` and in the frozen `dependencies/original/run_stage5.py`. These files establish implementation provenance, but the supplied records do not establish a first-principles derivation or an independently documented original selection rationale for $1/8$. We therefore retain it explicitly as a constitutive choice. It is not derived from the eight admission frames, and its successful scale is not evidence that eight frames are physically necessary.

Holding ρ_D / u_{crit} fixed gives $d \ln a_T = d \ln H_0 + (d \ln \kappa + d \ln(\rho_D / u_{\text{crit}}))/2$. An H_0 change of ± 0.5 around 67.4 changes the scale by $\pm 0.742\%$; a $\pm 1\%$ density-ratio change changes it by about $\pm 0.5\%$. These are conditional sensitivities, not a propagated cosmological posterior. Alternative coefficient scenarios with all other inputs fixed give:

κ	a_T in m/s^2	RAR RMS in dex
1/16	8.42721×10^{-11}	0.142060
1/8	1.19179×10^{-10}	0.132875
1/4	1.68544×10^{-10}	0.146099

This three-point sensitivity sweep is not an optimization or proof that $1/8$ is unique. It quantifies how much the descriptive comparison relies on that construction choice. The supplied empirical kernel is responsible for the response shape.

The rank-six minimality statement covers the complete class-resolved query family and arbitrary allowed class channels. It is not a proof that six blocks are minimal for replaying only the single fixed I/X update. The controlled X updates in Section 4 remain computational dynamics. A separate inherited fixed-volume check uses the same total positive energy operator as a dimensionless-clock unitary generator and verifies that sector exchange sums to zero (`state_load.py`). Neither check supplies a measured SI-time evolution law or derives cosmic expansion from microscopic dynamics.

6. Numerical controls and published-data comparison

All tests preserve the supplied SI coefficients. A dense 256-address, eight-dimensional random pure-state reference is evaluated directly and via three Σ_s . Ninety additional positive-operator queries give maximum error 2.9×10^{-15} . The full test has $N-1$ active addresses, $d=128$ and 32 fixed classes $n \bmod 32$ with seeded complex pure carrier states. Loads agree to 8.9×10^{-16} . Storing all dense conditional matrices would require 266,075,897,856 bytes in complex128; storing three weighted matrices needs 786,432 bytes. This is a theoretical storage comparison, not measured allocation. The grouped input already has a much smaller representation, and source construction still costs $O(N)$. No total-runtime speedup over all alternative implementations is claimed.

The same-marginals diagnostic uses two extremal eigenstates of $K_c/2$ and address groups $n \bmod 3=0$ versus the complement. Its paired correlated blocks have exactly the same carrier marginal as the product baseline and the same address distribution. Positivity is checked. The retained covariance changes D by -0.000406237 and R by $+0.035378346$; the error identity is satisfied to 3.1×10^{-16} . This is a constructed state test, not an observed cosmic correlation.

Diagnostic	Product marginals	Correlation retained
P load	0.050000	0.050000
D load	0.278926	0.278519
R load	0.687743	0.723121
q	-0.52856	-0.54501
Rotation, km/s	207.510	207.487
Conditional lens, arcsec	0.535582	0.535435

Twelve inherited shared-state configurations are reproduced by the scalar quotient with maximum printed-output discrepancy 2.9×10^{-14} . Equal-load microscopic alternatives therefore remain indistinguishable under this renderer, while the product-marginal ablation does not generally preserve those loads. These are different tests and support different conclusions.

The exact same-marginals construction is as follows. Let P_0, P_1 be the minimum/maximum eigenprojectors of $K_c/2$, $C=(P_0+P_1)/2$ and $D=(P_0-P_1)/2$. With $p=\sum_{n \bmod 3=0} w_n=0.2894395421456772$ and $t=\min(1, (1-p)/p)=1$, set $\rho_1=C+tD$ and $\rho_2=C-ptD/(1-p)$. Their weighted average is C and both are positive and trace one. The product baseline assigns C to every address. The original twelve inherited cases instead use I/d at every address, or a packet on the mod-3 class and I/d elsewhere; the packet is $(|8|+|16|+|24|)/\sqrt{3}$. They cover $\epsilon \in \{0,8\}$ and $a \in \{0.5,1,2\}$.

Rotation is evaluated at $r=8.2$ kpc for $g_b(r)=GM/(r^2+b_0^2)^{3/2}$, $M=6 \times 10^{10}$ solar masses and $b_0=3$ kpc. Lensing uses impact $b=10$ kpc, patch radius $R=200$ kpc and

$$\alpha(b) = \frac{4}{c^2} \int_0^{\sqrt{R^2-b^2}} g(\sqrt{b^2+z^2}) \frac{b}{\sqrt{b^2+z^2}} dz.$$

Radians are converted to arcseconds. This is a finite-patch weak-field diagnostic, not a measured lens system. The inherited twelve cases also compare independent 512-node Gauss–Legendre and adaptive quadratures, agreeing below 10^{-9} arcsec.

A fair representation comparison. For the 32-class complex pure-state diagnostic, retaining 32 vectors and the 3×32 weight matrix costs 66,304 bytes; three dense weighted operators cost 786,432 bytes; three scalar loads cost 24 bytes. The weighted representation is therefore larger than this grouped input. Ninety seeded, changing Hermitian carrier queries, evaluated for every sector, agree between grouped and weighted representations to 6.94×10^{-18} . The strengthened optimized workload sweep below evaluates practical utility including construction. Scalar loads cannot answer new carrier queries, while grouped states retain specified class-controlled information that a snapshot cache loses.

Workload sweep with construction included. The archived workload experiment tested $d \in \{32,64,128\}$, $G \in \{8,32,128\}$ and $m \in \{10,90,300\}$: 27 cases, five repetitions each, seeded complex pure states and positive $3 \times G$ weights. NumPy 2.3.5 uses OpenBLAS 0.3.30 on an AMD EPYC 9V74; OPENBLAS_NUM_THREADS, OMP_NUM_THREADS and MKL_NUM_THREADS are fixed to one before import. The maximum query discrepancy is 1.39×10^{-17} . Grouped queries use a direct BLAS matrix product followed by row contractions, with the conjugate buffer prepared once; contraction-path planning is not repeated. Reported retained bytes exclude working buffers for both methods. Query generation and address-weight compilation are common excluded inputs; cache construction is included. The table reports 90-query medians, in milliseconds. Full per-run variation is retained in workload_sweep_results.json.

d / G	Group / cache bytes	Group / cache total ms	Estimated break-even m
32 / 8	4,288 / 49,152	0.739 / 0.581	28
32 / 32	17,152 / 49,152	1.433 / 0.680	19
32 / 128	68,608 / 49,152	3.914 / 1.952	38
64 / 8	8,384 / 196,608	1.519 / 1.693	None
64 / 32	33,536 / 196,608	3.823 / 1.766	8
64 / 128	134,144 / 196,608	12.209 / 6.993	46
128 / 8	16,576 / 786,432	6.219 / 12.924	None
128 / 32	66,304 / 786,432	13.739 / 8.277	11
128 / 128	265,216 / 786,432	38.370 / 7.824	4

The cache including construction is faster in 16 of 27 local cases; it loses in 11 cases. Per-query saving and median build cost give estimated break-even counts from 4 to 49. This linear estimate is workload-dependent, not a promised crossover on other hardware. The five-run minimum and maximum values are archived rather than treated as inferential uncertainty. For this complex128-vector/float64-weight layout, $G < 3d^2/(d+1.5)$, the cache uses more

storage; for larger G it can use less. No single representation dominates. These measured costs operationalize the audit rather than identifying a physical minimality law.

Actual-interface failure and repair. Let $A_{\{sn\}} = w_{nf_s}(n)$ on the first 256 active addresses, $z_n = +1$ in the mod-3 class and -1 elsewhere, and $u_n = A_{\{Rn\}} z_n$. Define $h = u - A^T(AA^T)^{-1}Au$ and $\delta = h / \max |h_n|$. Set $p_n^{\pm} = (I \pm \delta_n Z) / 2$ on a carrier qubit for those addresses and $I/2$ elsewhere. All initial Σ_s coincide because $A\delta = 0$; all traces are one and positivity is explicit from $|\delta_n| \leq 1$. Apply I in the mod-3 class and X elsewhere. The difference of weighted Z queries becomes $2A \text{diag}(z)\delta$. The initial maximum difference is 1.12×10^{-16} ; after the update the R-sector difference is 0.6668017457457079 . This embeds in the 128-dimensional carrier. The six lifted operators from Proposition 3 retain the missing class information and replay the update. Thus the failure concerns all three WRRRA weighted operators, not merely an ordinary unweighted marginal. It gives a reproducible reason for retaining a larger state when moving from frozen outputs to address-controlled evolution.

7. Reproducibility and falsification

Current reproduction supplement (commit-pinned public archive): https://github.com/Wonsik-Choi-janefather/wrra-m-0.1/blob/923469bc980341ec91a98aed1b2b7ff7749d43a7/publication/WRRRA_3_0_v1_8/WRRRA_3_0_Reproduction_v1.8.tar.xz

SHA256: `b31a391c61e6843ef959a3e4ea82a8efa57cee6d9d2bdeb97478f2ffe7ff7dc1`. Download the archive and extract it with `tar -xJf WRRRA_3_0_Reproduction_v1.8.tar.xz`, then run the listed Python commands from the extracted directory. This v1.8 revision changes presentation and access documentation only; the model, numerical sources and reported results are unchanged from v1.7.

Original location of the inherited v1.4 archive:

https://github.com/Wonsik-Choi-janefather/wrra-m-0.1/blob/2f2b1c5e9e4c0625870fcddb5e2216737665e629/publication/WRRRA_3_0_v1_4/WRRRA_3_0_Reproduction_v1.4.zip. Archive SHA256: `ea30c242fd374a04f39b3562f76f7de26fd4659e75f18cf33d5a3cd39b79775b`. The adjacent manifest maps every included source and input to its hash.

The archive contains the full inherited sources for stages 1–7, their nested frozen dependencies and input manifests, the new benchmark, the official data bytes and a one-command reproduction entry point. The reference environment is Python 3.12, NumPy 2.3.5 and SciPy 1.17.0; Matplotlib is needed only for Figure 1. From the archive root:

```
python reproduce_all.py
python benchmark.py
python rar_comparison.py
python workload_sweep.py
```

The first command runs 27 stage scripts in an isolated copy and records exit codes, stdout, stderr and expected PASS conditions. The inherited execution record reports that all completed successfully; the current release retains that record. Stages use their pinned dependency snapshots: this is a full rerun of the published stage checks, not a newly refitted model. Different floating-point libraries can change final digits and timing measurements. The second and third commands recreate the new numerical controls and comparison from fixed inputs. `benchmark_results.json` and `rar_results.json` provide the full machine-readable values, including spectral error bounds. SHA256 checks cover the input data and distributed sources.

Verification has three distinct levels. The proved identities require their stated mathematical assumptions. Numerical disagreement principally tests software conformance or reveals a violated assumption. Observational residuals test the physical renderer against data; physical selection additionally requires a specified comparison and uncertainty model. The following failure conditions describe implementation conformance, not observational refutation of a proved identity. The exact interface fails if an independently implemented direct joint readout differs from its weighted-operator counterpart beyond the declared numerical tolerance, or positivity is lost. The marginal-error result fails if its covariance identity or spectral bound fails. The three-load equivalence fails if the same fixed coefficients and equal loads produce different downstream outputs. Conversely, if address-controlled updates are introduced, snapshot sufficiency alone is no guarantee; the counterexample is a test that rejects that extension. Supplier execution is rejected below its maximum branch requirement, and energy duplication or separate output refits invalidate the shared-ledger contract.

For observational claims, the frozen response can be compared to specified data and residual summaries; its association with measured acceleration is explicit here. A stronger physical selection test would require galaxy-level nuisance parameters and covariance, a fixed test set and a comparison of inequivalent microscopic constructions. No universal numerical rejection threshold is invented retrospectively from the present residuals. Current observational consistency and microscopic identifiability are separate claims.

Current release and claim mapping

Paper v1.8 and the current text of book v0.10 use the accompanying `WRRRA_3_0_Reproduction_v1.8.tar.xz`. It preserves the v1.4 source snapshots and adds the version map and `revision_audit.py`; the earlier URL identifies inherited provenance only. `RELEASE_MAP.json` maps this release, scripts and outputs, and `REPRO_MANIFEST_SHA256.json` lists file hashes. The current paper is the submission object. The previous cumulative book is preserved separately as historical documentation, not as additional current claims.

Current claim	Script from archive root	Output
Exact interface, covariance and rank-six repair	<code>benchmark.py</code>	<code>benchmark_results.json</code>
Conditional architecture comparison	<code>snapshots/source_step3/structure_comparison.py</code>	<code>adjacent structure_comparison_results.json</code>
Shared energy and macro states	<code>reproduce_all.py</code>	<code>rerun/FULL_RERUN.json</code> and stage outputs
RAR descriptive comparison	<code>rar_comparison.py</code>	<code>rar_results.json</code>
Optimized workload sweep	<code>workload_sweep.py</code>	<code>workload_sweep_results.json</code>
Calibration, kappa sensitivity, macro constraint	<code>revision_audit.py</code>	<code>revision_audit_results.json</code>

Run `python revision_audit.py` in addition to the commands above. The inherited 27-script stage rerun is retained with its original execution record. This revision reruns the benchmark, RAR comparison and new audit; historical timings are labeled as the previous measured workload rather than new measurements. No stage result is represented as an independently observed physical datum.

8. Conclusion

We have developed and audited a finite WRRRA toy model for exploring a shared explanatory chain across quantum-to-gravity, vacuum/expansion and dark-sector questions. The implemented contribution is the connection from finite SOURCE and common carrier through admission and phenotype to correlated loads and the stated energy, pressure and gravity readouts. The model uses disclosed calibrated physical inputs; its reproduction of known values is a consistency result. Exact readout reductions, the same-marginals counterexample and the class-controlled repair establish what information this implementation needs and what its present dynamics can support. The published-data comparison documents descriptive consistency of the frozen acceleration scale in a credited empirical kernel.

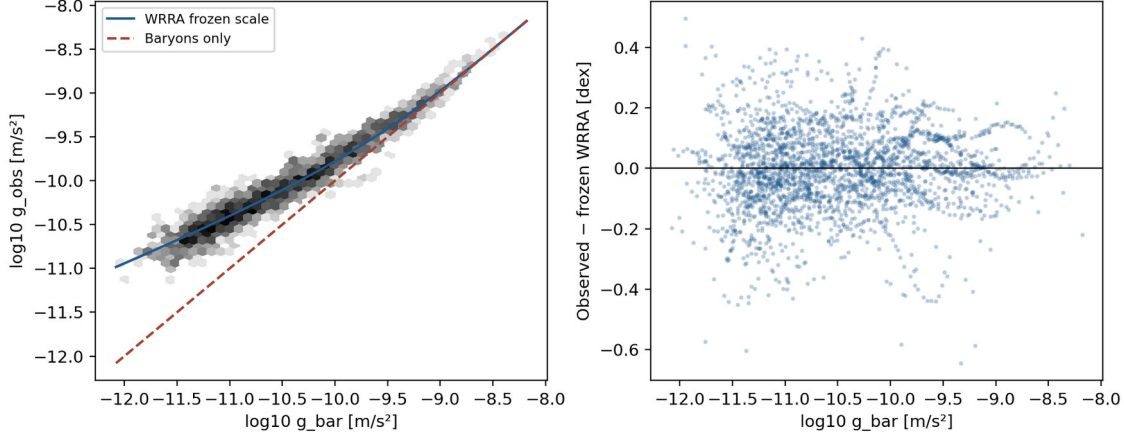
This model-development result is distinct from resolving the three physical problems. The microscopic quantum-gravity dynamics and vacuum-energy scale are not derived here, and the stated macroscopic readouts do not uniquely identify the arithmetic SOURCE. Those results define the model’s current explanatory reach and guide future extensions. The disclosed input–transformation–output chain and its failure conditions make the proposed model open to reproduction and further assessment.

Appendix A. Published-data consistency check

For an observational consistency comparison, we use all 2,693 rows of the official RAR.mrt release [5,6]. Its columns give $\log_{10}$ baryonic and observed acceleration and their marginal errors. The existing WRRRA scale $a_T = 1.191787531397112 \times 10^{-10} \text{ m/s}^2$ is fixed before this calculation. We change no stellar mass-to-light ratio, distance, inclination, cosmic fraction or response coefficient. The comparison is descriptive reuse of a previously published dataset and an inherited empirical function, not a blind validation or an independent discovery.

$$r_i = \log_{10} g_{obs,i} - \log_{10} \left[\frac{g_{bar,i}}{1 - e^{-\sqrt{g_{bar,i}/a_T}}} \right], RMS = \sqrt{\frac{1}{m} \sum_i r_i^2}.$$

Acceleration prescription	Scale, m/s ²	RMS residual, dex	Median residual, dex
Fixed WRRR state scale	1.19179×10^{-10}	0.132875	-0.001555
Published empirical RAR scale	1.20000×10^{-10}	0.132913	-0.002462
Baryons only	None	0.513981	0.430000



Published SPARC RAR points and fixed WRRR-scale residuals

Figure 1. Left: published points and fixed response curve. Right: observed-minus-calculated log residuals. The published empirical-scale curve differs from the WRRR-scale curve by at most 0.00143 dex and would almost overlap it. The RMS difference is not evidence of statistical superiority. Rows share galaxy-level nuisance parameters; this four-column release does not provide galaxy identifiers or a joint covariance. We therefore do not treat 2,693 rows as independent likelihood factors, quote a significance or estimate physical parameter uncertainty from this table. This result establishes descriptive consistency of the supplied scale and kernel with these data. It does not select the microscopic address law or validate conditional lensing.

Appendix B. Frozen supplier diagnostics

The three supplier branches are finite frozen inputs, not new particle constructions in this article. Their minimum budgets are maximum conditional branch-work requirements. At reference phenotype energy $E_P = 3.780791249536135 \times 10^{-11}$ J the retained values are:

Supplier branch	Mean work, J	Required budget, J
Effective record	$7.029276088428041 \times 10^{-12}$	$7.029707767554742 \times 10^{-12}$
Principal magnetic	$6.015266344305666 \times 10^{-12}$	$6.015700324113602 \times 10^{-12}$
Joint calibrated	$1.2033019830228727 \times 10^{-11}$	$1.2033440156388269 \times 10^{-11}$

The joint budget divided by E_P gives 31.8278354%. The first two suppliers are allocated budgets of 20% of E_P ($7.5615824991 \times 10^{-12}$ J); these allocations exceed the smaller required-budget thresholds listed in the table. The joint branch is allocated its disclosed required budget. Mean work alone is not sufficient for every conditional branch. `snapshots/source_step4/channel_record.py` and `mass_mixing_join.py` construct the three underlying branches; `snapshots/source_step5/record_load_join.py` tests their nested conservation. This paper takes these derived, pinned records as inputs and does not infer their microscopic origin from the conservation identity.

References

- [1] A. Jenčová and D. Petz. Sufficiency in quantum statistical inference. arXiv:math-ph/0412093. <https://arxiv.org/abs/math-ph/0412093>.
- [2] O. Kabernik, J. Pollack and A. Singh. Quantum State Reduction: Generalized Bipartitions from Algebras of Observables. Physical Review A 101, 032303 (2020). <https://doi.org/10.1103/PhysRevA.101.032303>.

- [3] T. Grigoletto, Y. Tao, F. Ticozzi and L. Viola. Exact Model Reduction for Continuous-Time Open Quantum Dynamics. *Quantum* 9, 1814 (2025). <https://doi.org/10.22331/q-2025-07-29-1814>.
- [4] S. S. McGaugh, F. Lelli and J. M. Schombert. The Radial Acceleration Relation in Rotationally Supported Galaxies. *Physical Review Letters* 117, 201101 (2016). <https://doi.org/10.1103/PhysRevLett.117.201101>; arXiv:1609.05917, Eq. 4.
- [5] F. Lelli, S. S. McGaugh and J. M. Schombert. SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. *Astronomical Journal* 152, 157 (2016). <https://doi.org/10.3847/0004-6256/152/6/157>.
- [6] F. Lelli, S. S. McGaugh, J. M. Schombert and M. S. Pawlowski. One Law to Rule Them All: The Radial Acceleration Relation of Galaxies. *Astrophysical Journal* 836, 152 (2017). <https://doi.org/10.3847/1538-4357/836/2/152>. Official published data: <https://astroweb.case.edu/SPARC/RAR.mrt>, retrieved 8 October 2026, SHA256 24aa7059dab7fa44787f7c11191052489899819370f6508621674769f3b72833.
- [7] W. Choi and J. Choi. Minimal Computing Cosmology 3.0, cumulative book and theory improvement stages. Previous public book release: <https://doi.org/10.5281/zenodo.23202732>. Frozen stage-six repository commit: <https://github.com/Wonsik-Choi-janefather/wrra-m-0.1/commit/dfa336c5526fe3e3923cee16e2a42dc2b3006412>. This revision includes snapshots for all preceding stages; no new DOI has been assigned to paper v1.8 or book v0.10.
- [8] Planck Collaboration. Planck 2018 results VI Cosmological parameters. *Astronomy & Astrophysics* 641, A6 (2020). <https://doi.org/10.1051/0004-6361/201833910>. Adopted reference H_0 under base-Lambda-CDM; project rounding conventions are stated separately.